CURRICULUM VITAE

Isaac W. Sluder

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EDUCATION

The University of Akron, Akron, Ohio (8/2024-7/2028)
Doctor of Philosophy in Engineering, Mechanical Engineering
Certificate, Computational Data Science, *in progress*
Current GPA: 4.00/4.00
Ph.D. Candidacy Completed May 2026
Advisor: Dr. K.T. Tan

The University of Akron, Akron, Ohio (8/2019-5/2024)
Bachelor of Science (Magna Cum Laude and Williams Honor Scholar)
Mechanical Engineering, Minor in Applied Mathematics
GPA: 3.771/4.000

Kent State University (branch), East Liverpool, Ohio (8/2014-5/2019)
College Credit Plus (CCP) Program
Participated in Ohio's College Credit Plus (CCP) program during high school, gaining early experience with college-level academics and STEM-focused coursework.

RESEARCH INTERESTS

Materials informatics, polymer and composite environmental durability, impact and dynamic response, knowledge graphs and scientific information extraction, machine learning for materials, data science, additive manufacturing.

RESEARCH EXPERIENCE

The University of Akron, Akron, OH
Graduate Research Assistant, Department of Mechanical Engineering (5/2024-Now)

- Conducted experiments to analyze the environmental effects of water and temperature on 3D printed composite materials.
- Architected a Large Language Model (LLM) workflow to aggregate environmental degradation data, streamlining the creation of a comprehensive database for predictive material modeling.
- Led experimental research on the hygrothermal degradation of additively manufactured chopped-carbon-fiber/polyamide composites, resulting in a first-author publication in *Composites Part B:*

Engineering.

Air Force Research Laboratory, Wright-Patterson AFB, OH
Summer Research Intern, Polymer Matrix Composites Branch, Foundational Technologies Directorate (5/2026-8/2026)

- Developed an end-to-end large language model pipeline for screening, extracting, normalizing, and organizing polymer-composite literature into a provenance-tracked Neo4j knowledge graph.
- Applied the system to a 1,452-document corpus on hygrothermal degradation of polyimides and bismaleimides
- Built a literature-grounded query system that generated interactive reports, data visualizations, and chemical structure diagrams with traceable source evidence.
- Reduced the estimated processing of a 1,500-paper corpus from 18 weeks of full-time chemist labor and approximately $35,000 in salary costs to 20 hours and approximately $140 in compute and API costs.

The University of Akron, Akron, OH
Undergraduate Research Student, Department of Mechanical Engineering (9/2023-5/2024)

- Executed comprehensive mechanical testing protocols on 3D printed carbon fiber reinforced polymers (CFRP), contributing to the advancement of materials science research.
- Initiated the mechanical recycling of CFRPs to evaluate the effects of reprocessing on material properties and sustainability.
- Facilitated weekly progress review meetings and presented findings and developments into concise presentations.
- Disseminated research results to the academic community by contributing to a manuscript and presenting at a local student conference.

The University of Akron, Akron, OH
Research Assistant, Department of Mechanical Engineering (6/2021-12/2021)

- Executed rigorous testing protocols on oils and greases, meticulously adhering to Standard Operating Procedures (SOPs) to maintain consistency and reliability.
- Employed data analysis techniques to extract valuable insights, subsequently optimizing processing times for enhanced efficiency and productivity.

INDUSTRY EXPERIENCE

M. K. Morse, Canton, OH
Mechanical Engineering Co-Op, R&D (1/2022-5/2024)

- Developed carbide tip orientation sorter utilizing machine learning vision technology, enhancing precision and efficiency in production processes.
- Engineered an embedded sensor system tailored for bandsaw back edge profiling, yielding research level value and optimizing back edge performance.
- Led proprietary research initiatives at the intersection of cutting-edge technologies and artificial intelligence, contributing to the company's strategic innovation roadmap.
- Innovated and executed designs for cutting tools and associated testing apparatus to support both research endeavors and product development initiatives.

The University of Akron, Akron, OH
Scanning Electron Microscope Manager, Akron Engineering Research Center (2/2023-8/2023)

- Orchestrated comprehensive training sessions to acquaint new users with SEM microscope functionalities, ensuring proficiency and adherence to best practices.
- Headed equipment management efforts, diligently addressing downtime and swiftly resolving technical issues to sustain uninterrupted workflow.
- Applied advanced testing methodologies to evaluate material properties essential for cutting-edge AI research, leveraging CNC scratch testing techniques to deliver precise and insightful data analysis.

Yellow Creek Casting Co., Wellsville, OH
Foundry Laborer (12/2018-5/2019)

- Specialized in mold-making within a foundry setting, proficiently utilizing green sand techniques and pouring molten iron into molds to craft intricate components.

TEACHING EXPERIENCE

The University of Akron, Akron, OH
Graduate Teaching Assistant, Department of Mechanical Engineering (8/2024-Now)

- Taught an undergraduate senior lab on mechanical pump systems, guiding students in hands-on experimentation and data analysis.
- Instructed a Concepts of Design lab, helping students develop a strong foundation in material selection and mechanical design principles.
- Designed and implemented a freshman Arduino & coding project to design a Bluetooth remote controlled vehicle, introducing students to embedded systems and hands-on programming.

HONORS & AWARDS

- **EDA Sustainable Polymers Tech Hub Workforce Initiative,** Sustainable Environment (WISE) grant (04/2026)
- **F3 RubberCity**, F3 HOF Leadership Award (04/2026)
- **Ohio Space Grant Consortium,** Summer Internship (06/2025 – 08/2025)
- **Ohio Space Grant Consortium,** Master's Fellowship (01/2025 – 05/2025, 08/2025 – 12/2025)
- **National Science Foundation,** National Research Traineeship (08/2025 – Now)
- **American Society for Composites,** 2025 Student Travel Award (10/2025)
- **American Society for Composites,** 2024 Student Travel Award (10/2024)
- **The University of Akron,** Mechanical Engineering Graduate Scholarship (08/2024 – Now)
- **The University of Akron,** Mechanical Engineering Under-Graduate Scholarship (2020)
- **The University of Akron**, Dean's List (08/2019 – 05/2020), (8/2021 – 5/2023)
- **The University of Akron**, President's List (08/2023)

PEER REVIEWED JOURNAL PUBLICATIONS

1. **Sluder, I. W.**, Frankowski, A. P., Tan, K. T., "Hygrothermal Degradation of Tensile and Impact Properties of 3D-Printed Chopped Carbon Fiber Composites", *Composites Part B: Engineering*, 327,

114107 (2026), 10.1016/j.compositesb.2026.114107

Manuscripts in Preparation/Submission

1. **Sluder, I. W.**, Frankowski, A. P., Cheng, E., Tan, K. T., "Matract: A Testbed for Evaluating Large Language Model–Driven Extraction Pipelines for Polymer Hygrothermal Degradation Knowledge Graphs", *Manuscript in preparation.*
2. **Sluder, I. W.**, Reinhart, G., Bezik, C., Simone D., Varshney, V., Tan, K. T., "A Large Language Model Pipeline for Curating Hygrothermal Degradation Data from Polyimide and Bismaleimide Literature", *Manuscript in preparation.*
3. Frankowski, A. P., **Sluder, I. W.**, Tan, K. T., "Mechanical Recycling and Additive Re-manufacturing of Short Carbon Fiber Thermoplastic Composites", *Submitted to Composites Part C: Open Access.*

CONFERENCE PUBLICATIONS & PRESENTATIONS

1. Sluder, I. W., Cheng, E., Tan, K. T., Large Language Model–Driven Automation of Hygrothermal Degradation Data Extraction for Thermoplastic Composites, **American Society for Composites 41st Technical Conference**, Sep 28-30, 2026, Madison, Wisconsin, USA
2. Sluder, I. W., Tan, K. T., Predicting Mechanical Degradation in 3D-Printed Nylon-Based CFRP from Environmental Factors, **American Society for Composites 40th Technical Conference**, Oct 5-8, 2025, Dayton, Ohio, USA
3. Sluder, I. W., Frankowski, A. P., Tan, K. T., Hygroscopicity of 3D Printed Carbon Fiber Reinforced Polymeric Composites, **American Society for Composites 39th Technical Conference**, Oct 21-24, 2024, San Diego, California, USA
4. Frankowski, A. P., Sluder, I. W., Tan, K. T., Mechanical Recycling of 3D Printed Carbon Fiber Reinforced Composites for Increased Sustainability Through a Reduction of Waste Material, **American Society for Composites 39th Technical Conference**, Oct 21-24, 2024, San Diego, California, USA
5. Ramachandran, M. K., Sumaiya, S. A., Golvaskar, M., Sluder, I. W., Rakurty, C. S., Rangasamy, N., Emuakpor, O. S., Kannan, M., Improving the Fatigue Life of an Additively Manufactured Stainless Steel Specimen Using a Secondary Grinding Process, **ASME Turbo Expo 2024 Conference**, June 25–27, 2024, London, England, United Kingdom
6. Sluder, I. W., Curtis, P. M., Tan, K. T., Curved Fiber Patterns in 3D-Printed Carbon Fiber Composites: A Flexural Study, **AIAA Region III 2024 Conference**, April 5-6, 2024, Akron, Ohio, USA
7. Frankowski, A. P., Sluder, I. W., Tan, K. T., Characterizing the IZOD Impact Strength of a 3D Printed Reinforced Carbon Fiber Composite Material with Varying Layup Orientations, **AIAA Region III 2024 Conference**, April 5-6, 2024, Akron, Ohio, USA

POSTER & RESEARCH PRESENTATIONS

1. Sluder, I. W. "Shattering the 20-Year Aerospace Material Discovery Loop: Human Machine Teaming to Bypass Months of Manual Review Towards Discovering Next-Gen Materials in Record Time." **Air Force Research Laboratory**, August 2026. Poster presentation.
2. Frankowski, A. P., Sluder, I. W., Curtis, P., Mack, J. P., Mirza, F., Tsai, I-C., Dong, Y., Tan, K. T. "Growing Up with GPT: Cultivating AI Fluency and Ethics in K–12 Student Teams." **NSF National Research Traineeship poster presentation**, Spring 2026.

3. Sluder, I. W. "Extreme Environmental Effects on 3D Printed Composites for Space Applications." **Ohio Space Grant Consortium**, Summer 2025. Oral presentation.

PROFESSIONAL SERVICE

- **Manuscript/Conference Reviewer**, American Society for Composites 40th Technical Conference, 2026
- **Manuscript Reviewer**, International Journal of Lightweight Materials and Manufacture, 2026

VOLUNTEER & OUTREACH ACTIVITIES

- **The Chapel**, Worship Team Leader, and Volunteer (05/2025-Now)
- **F3 Akron**, Site Little-Q (07/2026–Now)
- **F3 Akron**, Volunteer Leader (Q) (05/2025–Now)
- **Presidential AI Challenge** – Hudson City Schools, Coach & Mentor (09/2025–01/2026) — Mentored elementary students in developing GooseBot: An AI-Powered Solution for Cleaner and Safer Community Parks, Ohio Track I State Winner (2026)
- **The Chapel**, Campus Focus, Worship Team Leader, and Volunteer (08/2019-05/2025)
- **The Chapel**, Campus Focus, Small Group Leader (08/2020-05/2024)
- **Science Olympiad,** Grader (03/2024)
- **The Chapel**, Trunk or Treat Event (10/2022, 10/2023, 10/2025)
- **Haven of Rest Ministries,** Volunteer (2019)

MEMBERSHIP IN PROFESSIONAL ORGANIZATIONS

1. American Society for Composites (ASC) 2024–Now
2. American Institute for Aeronautics and Astronautics (AIAA) 2024–2025

TECHNICAL SKILLS

Computational Skills
Programming: Python, MATLAB, R, C++, Julia
Data & Databases: Neo4j/Cypher, SQLite, InfluxDB, pandas
AI/ML & Cloud: Google Cloud, Vertex AI, LLM APIs, Langflow
Engineering Software: SolidWorks, Fusion 360, OriginPro, Minitab
Developer Tools: Git/GitHub, Codex, Antigravity, VS Code, Jupyter, Bash/Zsh/PowerShell

Experimental Techniques:
Mechanical Testing: Dynamometer (Kistler), Instron 5582 Universal Testing Machine, Instron CEAST 9350 impact testing
Microscopy & Imaging: μCT imaging, SEM – Tescan Lyra 3 XMU with EDAX, Keyence VHX-7000, Zygo Microscope
Tribology: Penetrometer, MPR – PCS Instruments, High-Frequency Reciprocating Rig, Ultra-High Pressure Viscometer, Rheometer
Additional Techniques: Additive Manufacturing Machines, Environmental Chambers

PROFESSIONAL REFERENCES

- **Dr. Kwek-Tze "K.T." Tan**
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